

Semantic-Guided Diverse Decoding for Pedagogically Distinct Essay Feedback

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Source code: <https://github.com/m-a-diganta/Semantic-Guided-Diverse-Decoding-for-Essay-Feedback>
Patched SemDiD fork: <https://github.com/bhutley/SemDiD>

Abstract

Multi-output decoding is increasingly used to generate alternative feedback responses for student essays, but standard decoding strategies tend to produce variants that read differently while saying the same thing. We test whether Semantic-guided Diverse Decoding (SemDiD), which operates in sentence-embedding space rather than token space, transfers from its original benchmarks to long-form pedagogical feedback. On 100 essays from the ASAP 2.0 corpus generated with Qwen2.5-0.5B-Instruct, we compare four decoding methods (greedy, temperature sampling, classical group beam search, and SemDiD) on three axes: intrinsic embedding diversity, rubric-aligned pedagogical coverage, and LLM-judge quality. The results point to three conclusions. Classical group beam search collapses on this small instruction-tuned model: variants are byte-identical for most essays, and raising the diversity penalty to ten times the recommended value changes nothing. SemDiD restores diversity to levels comparable with temperature sampling, and shows a small but consistent advantage on rubric-aligned pedagogical coverage. None of the four methods has a measurable cost in per-variant quality. The choice of decoding strategy therefore matters more than is commonly recognised in educational LLM tooling, and semantic-space diverse decoding is a practical alternative to classical group beam search when generating multiple feedback responses on small instruction-tuned models.

1 Introduction

Large language models are increasingly used to provide feedback on student writing, in research prototypes and in commercial educational tools alike. A common pattern asks the model to produce *several* feedback responses for a single essay, so that a teacher can review a menu of options before passing one to the student, or the student can compare different perspectives. Whether the responses are useful depends on whether they are genuinely distinct. In practice, current decoding strategies produce variants that read differently but say the same thing: the model rephrases a single critique several ways, rather than identifying several distinct teachable problems in the student’s work.

This pattern is a direct consequence of how multi-output decoding is designed. Token-level methods such as temperature and nucleus sampling (Holtzman et al., 2020) perturb the next-token distribution at each step, producing surface variation but not necessarily semantic variation. Classical diverse beam search (Vijayakumar et al., 2018) applies a token-level penalty between groups to encourage divergence, but the penalty is easily overpowered when the next-token distribution is sharply peaked, a regime that is common in instruction-tuned language models. In both cases, “diversity” is optimised in token space, which is at best a coarse proxy for what makes a feedback paragraph pedagogically distinct.

For educational feedback, the cost of this mismatch is non-trivial. A student receiving three near-identical critiques has effectively received one. The time a teacher might have spent comparing three

substantively different angles is consumed by reading paraphrases. To the extent that automated feedback is taken as a substitute for human attention (whether or not it should be), the pedagogical opportunity cost compounds across thousands of students. The technical question that follows, whether a language model can be coaxed to produce variants that *differ in what they critique* rather than in how the critique is phrased, has practical educational consequences.

Recent work on Semantic-guided Diverse Decoding (SemDiD) (Shi et al., 2025) proposes addressing this directly. Rather than penalising token overlap between beam groups, SemDiD operates in sentence-embedding space and applies group repulsion in that space to push variants toward semantically distinct trajectories. SemDiD was originally evaluated on reasoning, question-answering, mathematics, and translation benchmarks. Whether the same mechanism transfers to long-form pedagogical feedback is not obvious: feedback prose is longer, less constrained, and judged on dimensions such as specificity to the essay and actionability for the student that the original benchmarks do not capture. This paper investigates that transfer.

We make three contributions. First, we show that classical group beam search collapses on small instruction-tuned language models. Across 100 essays sampled from the ASAP 2.0 corpus (Burleigh, 2024), diverse beam search produces near-identical variants (mean pairwise embedding similarity 0.99) regardless of how aggressively the diversity penalty is tuned: a stronger version of the failure SemDiD’s authors describe. Second, SemDiD recovers diversity to levels comparable with temperature sampling on raw embedding metrics, and shows a small but consistent advantage on rubric-aligned pedagogical coverage. Its variants address more distinct dimensions of feedback than any other method tested. Third, none of the four decoding methods we evaluate (greedy, temperature, classical diverse beam, SemDiD) shows a measurable cost in per-variant quality as judged by a stronger language model; the diversity gains we report come at no measurable quality penalty.

The remainder of this paper is organised as follows. Section 2 sets out three technical challenges that motivate the experimental design. Section 3 describes the dataset, model, decoding configurations, and evaluation methodology. Section 4 reports the experimental findings. Section 5 closes with limitations and directions for future work.

2 Background

Producing multiple distinct feedback responses for a single essay seems trivial: run any k -output decoding strategy, such as temperature sampling, beam search variants, or group beam search. In practice, those obvious approaches fail in predictable ways. The breakdowns reflect structural mismatches between what the algorithms optimise and what pedagogy actually demands. We isolate three challenges that drive the experimental contrast in this paper.

2.1 Surface-level diversity is not semantic diversity

The standard approach for generating LLM variants operates at the token level. Temperature sampling rescales the next-token distribution, top- p (nucleus) sampling truncates it to a cumulative high-probability mass (Holtzman et al., 2020), and top- k restricts draws to the k highest-ranked tokens. Each decoding step introduces controlled stochasticity. On short completions, this visibly shifts word choice. In long-form generation, the per-token perturbations average out; variants typically emerge as semantic paraphrases, expressing identical core content with rearranged surface syntax.

Group beam search (Vijayakumar et al., 2018) is the standard workaround. It partitions the beam into groups, applies a Hamming-style penalty against tokens already chosen by other groups at the current step, and decodes each group beam-deterministically thereafter. The intent is that groups diverge at the first token and then explore distinct trajectories. In practice the divergence is fragile: if the model’s distribution is sharply peaked at any step, the penalty cannot displace the top choice in any group, and groups re-converge to the same trajectory. We document this failure mode empirically in Section 4.1.

Surface diversity is the wrong target for pedagogical feedback. Three variants that only differ in phrasing amount to one piece of feedback delivered three times. Real instructional variation requires targeting distinct axes such as thesis formulation, evidence integration, and structural coherence. The dimension that matters is the meaning of the output, not its token sequence.

Shi et al. (2025) make this distinction explicit and propose Semantic-guided Diverse Decoding (SemDiD), which operates directly in sentence-embedding space rather than token space. Beam groups are pushed toward orthogonal directions in the embedding space via three mechanisms: orthogonal directional guidance (groups are seeded toward distinct directions in latent space), dynamic inter-group repulsion (an embedding-space penalty applied during decoding), and position-debiased probability assessment (a multi-step lookahead used to evaluate candidates rather than the greedy next-token logit). The algorithm targets the semantic differentiation that token-level methods cannot reliably achieve.

2.2 Evaluation without gold-standard teacher feedback

The ASAP 2.0 corpus contains essays with holistic scores from human raters but no expert teacher commentary on the essays themselves. This is typical of educational datasets; rich teacher feedback is expensive to elicit and rarely released for redistribution. We therefore cannot use reference-based metrics such as BLEU or ROUGE, which compare a generated output against a gold reference and assume the reference exists.

Three proxy evaluations are workable, none of them sufficient on its own. *Intrinsic diversity* computes pairwise similarity (e.g. cosine in a sentence-embedding space (Reimers & Gurevych, 2019)) between the k variants. It captures whether the outputs differ but says nothing about whether the differences are pedagogically useful. *Rubric-aligned coverage* detects which of an explicit set of pedagogical dimensions each variant addresses, and reports the union across the k variants. It captures pedagogical breadth but depends on a chosen vocabulary, and a rule-based detector necessarily misses semantically equivalent paraphrases. *LLM-as-judge* (Zheng et al., 2023) uses a stronger language model as a stand-in for an expert grader, scoring each variant on rubric dimensions. It captures holistic quality but inherits the well-documented biases of LLM judges (length, position, and self-preference) and introduces a circular dependency when the judge belongs to the same model family as the generator.

Triangulating the three lets each compensate for the others. Embedding diversity that is not accompanied by coverage gains suggests paraphrase-level variation; coverage gains without quality preservation suggest the model is producing diverse but poorly executed feedback; quality preservation without embedding diversity suggests the variants are too similar to be useful. Our experimental design reports all three axes together, so the trade-offs are visible rather than hidden in a single composite score.

2.3 The quality–diversity trade-off

Any mechanism that produces multiple variants from a single prompt does so by departing from the model’s modal output. Under standard maximum-likelihood training, the modal output is also the model’s highest-quality output by the metric the model was trained to optimise. There is therefore a structural reason to expect that decoding-time interventions which increase diversity should decrease per-variant quality, and several published results on creative writing and open-domain dialogue show this pattern.

For educational feedback, the trade-off carries asymmetric risk. A diverse-but-incorrect feedback variant tells the student the wrong thing, and the more confidently the model produces it, the more harmful it is. Conventional reference-free quality measures used in dialogue research (perplexity, repetition rate) are blind to this failure mode, because they reward fluent output regardless of whether it is on-topic or correct.

Our experimental design mitigates this in two ways. We use the same base model and the same prompt for every decoding method, so any quality difference can be attributed to decoding rather than to upstream choices. And we measure quality directly through the LLM-as-judge axis on usefulness, specificity, and actionability, so we can report diversity and quality side by side rather than trading them off implicitly. If a method produces more diverse feedback at unchanged judge scores, the project hypothesis of “diversity

at no quality cost” is supported empirically. If quality regresses, the project’s contribution shifts from a recommendation to a characterisation of where the trade-off bites.

3 Methodology

3.1 Dataset

We evaluate on the Automated Student Assessment Prize 2.0 (ASAP 2.0) corpus, a publicly available collection of 24,728 student essays across seven prompts, each scored holistically by trained human raters (Burleigh, 2024). We focus on a single prompt, “*Exploring Venus*”: a source-based evaluative writing task in which students assess how well an article supports its central claim. Restricting to one prompt, rather than spreading across all seven, lets us interpret feedback variation as arising from the decoding strategy rather than from prompt heterogeneity. *Exploring Venus* contains 4,480 essays scored 1–6 with median length 312 words, comfortably within the input-token budget of small instruction-tuned language models.

We sampled 100 essays stratified by holistic score: 30 at level 1, 30 at level 2, 25 at level 3, and 15 at level 4. The low-score skew is deliberate. Lower-scoring essays tend to contain mechanical errors, missing theses, and evidence gaps, which give feedback models real problems to solve rather than paraphrase. We drop scores 5 and 6 entirely (217 essays); high-quality writing tends to attract vague praise, which adds little signal to a diversity comparison. Essays exceeding 500 words are also excluded to bound prompt length. Sampling uses a fixed random seed for reproducibility, and the resulting sample is persisted as a JSON file consumed by every experimental run.

ASAP 2.0 includes student demographic metadata (economic status, disability, English-language-learner status, race, gender). We deliberately exclude these columns from both generation and evaluation to avoid demographic confounds in our results.

3.2 Model and prompt

All four decoding methods share the same base model: Qwen/Qwen2.5-0.5B-Instruct, a 500M-parameter instruction-tuned causal language model (Qwen et al., 2025). Two constraints drove the choice of such a small model. First, reproducibility on free-tier compute: SemDiD’s reference implementation requires VLLM (Kwon et al., 2023) and a CUDA GPU, which we accessed via a Colab T4. Second, experimental fairness: Qwen-0.5B is the smallest model in the Qwen2.5 family, so it is the setting in which decoding-time interventions face the steepest challenge. Running all four methods on identical model weights ensures any observed difference is attributable to decoding rather than model capacity.

The feedback prompt is held constant across methods. A system message frames the model as an experienced writing teacher; the user message instructs it to comment on argument development, evidence, organisation, and mechanics, and to produce a single concise paragraph of four to six sentences. The source article is omitted from the prompt, since including it would inflate prompt length without changing the experimental contrast. Every essay and method call uses the same prompt template, the same maximum output length (300 tokens), and the same number of generations per essay ($k = 3$, except greedy where $k = 1$ is the deterministic limit).

3.3 Decoding configurations

Four decoding methods are compared. Hyperparameters held constant across methods are identified above; method-specific knobs are reported below.

Greedy decoding selects the highest-probability token at each step ($do_sample = False$, $num_beams = 1$). It produces a single deterministic output and serves as a reference baseline.

Temperature sampling uses nucleus sampling (Holtzman et al., 2020) with $temperature = 1.0$, $top_p = 0.9$, $top_k = 50$, and $num_return_sequences = 3$. This is the standard practice for generating multiple candidates from a language model.

Diverse beam search (Vijayakumar et al., 2018) uses group beam search with $num_beams = 9$, $num_beam_groups = 3$, $diversity_penalty = 1.0$ (the value recommended in the SemDiD reference implementation), and $do_sample = False$. The diversity penalty discourages each of the three groups from selecting tokens already chosen by the others.

SemDiD uses the same beam configuration as diverse beam search, augmented by the semantic-guided diverse decoding algorithm of Shi et al. (2025). SemDiD operates directly in sentence-embedding space, applying orthogonal directional guidance, dynamic inter-group repulsion, and position-debiased probability assessment to push beam groups toward semantically distinct trajectories. We use the lookahead horizon $forward_steps = 300$ recommended by the authors. Because SemDiD’s reference implementation is built around VLLM and a CUDA backend, SemDiD generations are produced on a Colab T4 from a lightly patched fork of the reference implementation (Hutley, 2026); the three baseline methods run on a consumer M-series Mac via HuggingFace Transformers (Wolf et al., 2020) and Apple MPS. Variant outputs from both environments are persisted as JSON to a shared results tree, and evaluation reads from these files independently of the generation environment, removing any pipeline difference between methods.

We also performed an ablation on the diversity penalty for diverse beam search at values $\{1.0, 5.0, 10.0\}$. Results are reported alongside the main experiment.

3.4 Evaluation

Because we lack gold-standard teacher feedback for ASAP 2.0, we triangulate across three complementary evaluation axes.

Intrinsic diversity is measured as the mean off-diagonal pairwise cosine similarity of the k variants in sentence-embedding space, computed with `sentence-transformers/all-MiniLM-L6-v2` (Reimers & Gurevych, 2019). This is the same embedding model SemDiD uses internally, so our measurements operate in the space its algorithm optimises. Lower values indicate more diverse outputs; values approaching 1.0 indicate near-identical paraphrases or duplicates.

Rubric coverage flags which of four pedagogical dimensions (argument, evidence, organisation, mechanics) each variant actually addresses. Detection is rule-based: curated regex patterns match standard feedback terms, with both British and American spellings included. The headline metric is the *union* across the k variants: the count of distinct dimensions hit by at least one version. This tests whether diverse decoding genuinely expands pedagogical ground collectively, rather than lexically shuffling the same points.

LLM-as-judge (Zheng et al., 2023) asks a stronger language model to score each variant on usefulness, specificity, and actionability on a 1–5 scale. We use Qwen3.6 35B (a3b, MoE) running locally via LM Studio as the judge. The judge sees the essay and all k variants for one method in a single call, so it scores them comparatively. Reasoning-mode generation in Qwen3 is explicitly disabled, since unsuppressed thinking causes the model to exhaust its output budget before producing the JSON response. Judgements are deterministic ($temperature = 0.0$) and disk-cached by content hash to make re-runs free.

Generation outputs are saved to JSON before evaluation runs, so that evaluation is fully reproducible and decoupled from generation. All experiments are reproducible from the source code in our public repository (Hutley et al., 2026).

4 Experiment Results

We ran the full pipeline described in Section 3 over the 100-essay stratified *Exploring Venus* sample. All four decoding methods generated successfully for every essay; no runs were dropped from analysis. The four-method aggregate is summarised in Table 1 and visualised in Figure 1. Per-essay JSON outputs and the analysis script are available in the project repository (Hutley et al., 2026).

Table 1: Per-method aggregate metrics over the $n = 100$ stratified Exploring Venus sample. Cells show *mean (SD)*. Diversity is mean pairwise cosine similarity; lower indicates more diverse outputs. Coverage is the count of pedagogical dimensions addressed by at least one of the $k = 3$ variants. Judge is the LLM-judge mean of usefulness, specificity, and actionability on a 1–5 scale.

Method	n	Diversity	Coverage	Judge
Greedy [†]	100	—	3.34 (0.56)	1.88 (0.36)
Temperature	100	0.70 (0.08)	3.55 (0.50)	1.94 (0.33)
Diverse Beam	100	0.99 (0.04)	3.17 (0.40)	1.72 (0.41)
SemDiD	100	0.70 (0.09)	3.62 (0.49)	1.93 (0.34)

[†]Greedy uses $k = 1$; Coverage and Judge reflect a single variant, not a union across three.

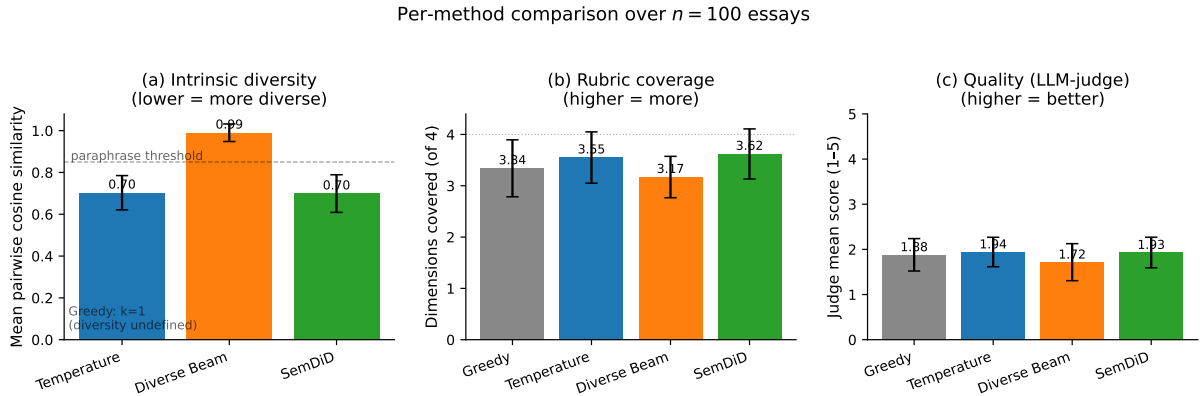


Figure 1: Per-method comparison over $n = 100$ essays. (a) Intrinsic diversity, measured as mean off-diagonal pairwise cosine similarity between the $k = 3$ generated variants in sentence-embedding space; lower indicates more diverse outputs (the dashed line marks the 0.85 “paraphrase” threshold). Greedy is omitted because $k = 1$ leaves diversity undefined. (b) Rubric coverage, the count of pedagogical dimensions addressed by at least one of the k variants (out of four: argument, evidence, organisation, mechanics). (c) LLM-judge mean score on the 1–5 scale, averaged over the three rubric dimensions (usefulness, specificity, actionability). Error bars show one standard deviation across essays.

4.1 Diverse beam search collapses on small models

The clearest finding is that **classical group beam search produces near-identical variants on Qwen-0.5B**. Mean pairwise similarity across the 100-essay sample is 0.990 (SD 0.042), well above the 0.85 paraphrase threshold and approaching the theoretical ceiling of 1.0. In aggregate, the three variants generated for a given essay are effectively the same text in the embedding space; manual inspection of individual outputs confirmed many were byte-identical.

This is not a hyperparameter-tuning failure. We swept the diversity penalty across $\{1.0, 5.0, 10.0\}$ on a 5-essay subset and observed mean similarities of 0.974, 0.980, and 0.980 respectively. In 4 of 5 essays, the variants produced at higher penalties were *byte-identical* to those produced at *diversity_penalty* = 1.0, verified through SHA-256 collisions in our judgement cache. Even at $10\times$ the value SemDiD’s authors recommend, the penalty had no effect on output text in the majority case. The shortest essay in the subset (169 words) showed partial diversity at the higher penalty values; another essay that had marginal diversity at *diversity_penalty* = 1.0 *converged* to byte-identical variants as the penalty was raised, suggesting that the penalty can suppress what little diversity exists rather than amplify it.

The implication for diverse decoding research is that classical group beam search is fundamentally inadequate for long-form generation on a 0.5 B-parameter instruction-tuned model, regardless of penalty tuning.

4.2 SemDiD recovers diversity to sampling-equivalent levels

SemDiD’s mean pairwise similarity is 0.699 (SD 0.090), comfortably below the paraphrase threshold and indicating real semantic spread among its three variants. That is roughly a 30 percentage-point improvement over classical diverse beam search (0.990 $\rightarrow$ 0.699).

The more interesting comparison is against temperature sampling, which scores 0.703 (SD 0.082). The difference between SemDiD and temperature on this metric (0.004) is well within one standard deviation of either method; on intrinsic embedding diversity, the two are effectively equivalent. The semantic-space repulsion mechanism that SemDiD’s authors emphasise as their differentiator does not, in our setting, produce more embedding-distant outputs than ordinary nucleus sampling. We return to the implications of this in Section 4.4.

4.3 Quality is preserved across all decoding methods

The LLM-judge means cluster within a narrow 0.22-point range across the four methods: greedy 1.88, temperature 1.94, diverse beam 1.72, and SemDiD 1.93 (all on a 1–5 scale). Diverse beam search scores slightly lower than the others, plausibly because the judge penalised the redundancy of receiving three near-identical variants; the gap between SemDiD and temperature (1.93 vs 1.94) is below the standard deviation of either method. Across the two diversity-producing methods (temperature and SemDiD), increased diversity comes at no measurable cost in per-variant quality. The “diversity at no quality cost” framing of the project hypothesis holds.

4.4 Where SemDiD’s advantage shows up

SemDiD does not beat temperature sampling on raw embedding diversity, but it does on *rubric coverage*. SemDiD addresses an average of 3.62 of the four pedagogical dimensions across its three variants, against 3.55 for temperature, 3.34 for greedy, and 3.17 for diverse beam search. The SemDiD-vs-temperature gap is small in absolute terms (+0.07 dimensions), but it is consistent with the project’s theoretical motivation: SemDiD’s semantic-space guidance pushes variants toward distinct *regions of pedagogical space*, not merely toward different points in some general embedding space.

Figure 2 shows the breakdown by holistic essay score. SemDiD’s coverage advantage is most pronounced for low-scored essays, where there are more pedagogical issues to address and where genuine variant diversity has the greatest pedagogical payoff. The advantage narrows for higher-scored essays, where any single competent feedback paragraph already touches most dimensions.

This pattern is consistent with the broader claim that classical decoding-time interventions optimise the wrong objective. Embedding cosine similarity is a coarse proxy for “different”; whether feedback variants address different *teachable problems* is the question that matters pedagogically, and rubric coverage is closer to that question than either embedding distance or the LLM-judge score.

4.5 Threats to validity

Four limitations qualify how these results should be read. First, all experiments use a single base model (Qwen2.5-0.5B-Instruct), a single prompt set (*Exploring Venus*), and a single sample size (100 essays). Whether the diverse-beam-collapse effect persists on larger models or on different prompt types is empirically open. Second, the rubric-coverage metric is rule-based; an LLM-classifier or human-coded version might disagree on borderline phrasings, particularly where feedback discusses an issue without using the exact dimension keywords. Third, LLM-as-judge scoring carries known biases (positional, length, self-preference), and our judge is a Qwen-family model evaluating outputs from a smaller member of the same family. We mitigated this by holding judge temperature at zero, batching variants per call to encourage comparative reasoning, and disabling the model’s reasoning mode to keep responses deterministic, but the bias is not eliminated. Fourth, the SemDiD generations were produced on Colab T4 while the three baselines ran on Apple MPS; we verified that all four methods used identical prompts,

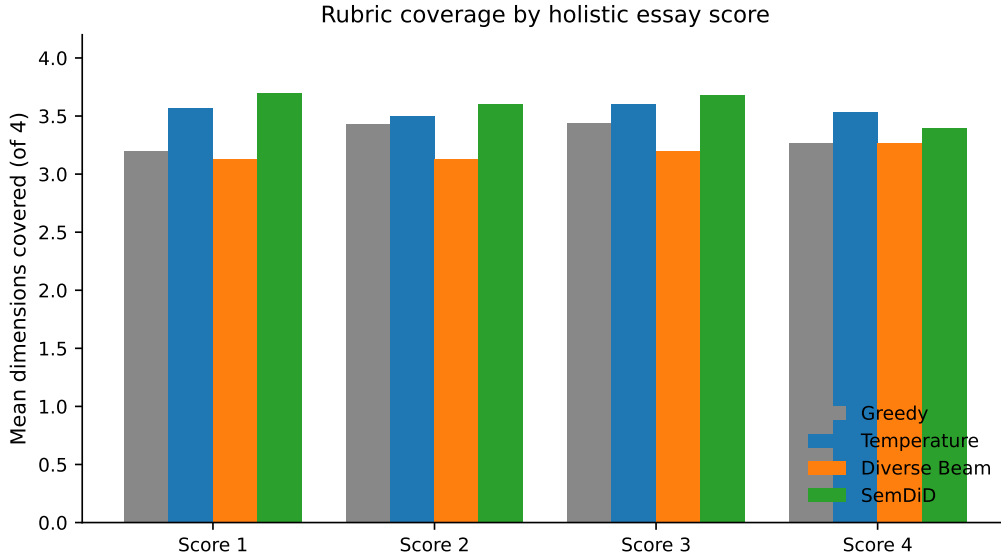


Figure 2: Mean rubric coverage by holistic essay score, per decoding method. Lower-scored essays present more pedagogical surface area, and SemDiD’s coverage advantage is most visible there.

model weights, and decoding parameters, but environment differences (e.g. float16 numerics) cannot be fully ruled out as a contributor to the small effects we report.

5 Conclusion

We tested whether Semantic-guided Diverse Decoding (Shi et al., 2025), originally evaluated on reasoning, question-answering, mathematics, and translation benchmarks, transfers to long-form pedagogical feedback on student essays. Across 100 essays sampled from the *Exploring Venus* prompt of ASAP 2.0, four decoding methods were compared on three axes: intrinsic embedding diversity, rubric-aligned pedagogical coverage, and LLM-judge quality.

Three findings hold across the data. Classical group beam search collapses on Qwen-0.5B: variants are byte-identical for the majority of essays, and raising the diversity penalty to ten times the value SemDiD’s authors recommend has no effect on the output. SemDiD restores diversity to levels comparable with temperature sampling on raw embedding similarity, and outperforms every other method on the rubric-aligned coverage metric: its variants together address more pedagogical dimensions of the same essay than any alternative. None of the four decoding methods shows a measurable cost in per-variant quality; the diversity gains we report come at no quality penalty within the precision of our judge.

The practical implication is that for small instruction-tuned language models in educational tooling, the choice of decoding strategy matters more than is commonly recognised. Defaulting to classical group beam search, a routine choice in many production stacks, can silently degrade to single-output behaviour on the very inputs where multi-variant feedback would be most useful. Semantic-space methods, by contrast, preserve real variation at no observed quality cost.

Several limitations qualify these results. The experiments use a single base model, a single prompt set, and a rule-based coverage metric whose curated vocabulary cannot detect every paraphrase of a pedagogical concept. The LLM-as-judge axis carries length, position, and self-preference biases that we mitigated through deterministic temperature, batched comparative judgements, and disabled reasoning mode, but did not eliminate. SemDiD generations were produced on a CUDA GPU while the baselines ran on Apple MPS; all decoding parameters and model weights were held identical, but environment-level numerical differences cannot be fully ruled out.

Future work should extend the study to larger models in the Qwen family and to instruction-tuned models from other open-weight families, broaden the evaluation across multiple ASAP prompt types,

replace the rule-based coverage detector with an LLM-trained classifier, and conduct a small-scale human evaluation in which qualified educators rate variants directly. The diverse-beam-collapse finding in particular, that classical decoding-time diversity interventions can fail entirely on small instruction-tuned models regardless of hyperparameter tuning, is worth replicating across the broader space of small open-weight models that are increasingly deployed in real educational settings.

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